

Abhishek Singh

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EDUCATION

- **Rochester Institute of Technology(RIT), Rochester** NY, USA
Master in Data Science; GPA: 3.9 Aug. 2023 – May. 2026

EXPERIENCE

- **Morningstar** Mumbai, India
Associate Software Engineer May 2021 - May 2023
 - **Frontend Engineering:** Built and enhanced portfolio analytics tools in Vue.js, improving report clarity and increasing automated test coverage across production components.
 - **Content Platform Architecture:** Designed CMS-driven FAQ system using Contentstack APIs, enabling non-technical stakeholders to manage documentation independently.

PUBLICATIONS

- **Singh, A.,** Lyu, Z., Desell, T. “Biologically-Inspired Homeostasis for Neuroevolution: Alternating Growth and Pruning Phases.” Accepted at *EvoStar 2026*.
 - **Research Contribution:** Reduced evolved neural model size by 23% ($103.1 \rightarrow 79.0$ parameters, $p = 0.0019$) without degrading validation MSE using phase-controlled growth-pruning schedules in distributed neuroevolution for **time-series forecasting**.

PROJECTS

- **AI Travel Itinerary Planner**
LangChain, Kubernetes, GCP, ELK
 - **Agentic Orchestration & Scalable Deployment:** Built a LangChain-based, real-time itinerary orchestration system with containerized microservices on GCP Kubernetes, enabling high-availability LLM inference and autoscaling for production workloads.
 - **Observability & LLMOps:** Implemented an ELK-driven telemetry and LLMOps pipeline to monitor token usage, query latency, and model performance, enabling rapid iteration, debugging, and operational reliability.
- **Generative AI Video Compliance System**
LangGraph, Azure Video Indexer, Azure AI Search
 - **Agentic RAG Pipeline:** Built multi-modal compliance system combining OCR/transcripts with regulatory retrieval using Azure AI Search and LangGraph orchestration.
 - **LLMOps & Observability:** Integrated LangSmith tracing and Azure Monitor telemetry for prompt evaluation and latency optimization.
- **Multi-Agent Quantitative Analysis System**
CrewAI, OpenAI API, FastAPI, Azure
 - **Autonomous Financial Agents:** Engineered multi-agent system for market scraping, trend analysis, and automated report generation.
 - **Scalable Backend Architecture:** Developed asynchronous FastAPI backend with Azure PostgreSQL and Blob Storage for structured and unstructured financial data.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C++, Java, JavaScript, Bash
- **AI & Machine Learning:** Distributed Computing, PyTorch, TensorFlow, LLMs (Fine-tuning, Agents), RAG, LangChain, LangGraph, **Azure OpenAI**, NetworkX, Transformers (Hugging Face), XGBoost, Scikit-learn
- **Infrastructure & Cloud:** **Azure (Video Indexer, Monitor, AI Search)**, AWS (Bedrock, Lambda, S3), **FastAPI, OpenTelemetry**, Docker, Kubernetes, HPC (Slurm, MPI), Linux, Git
- **Data & Databases:** PostgreSQL, SQLite, **Azure AI Search (Vector Store)**, Pinecone, **Pydantic**, Pandas, NumPy, Matplotlib, Airflow